

Paper 12 §3 — The NP-Pump Nonlinearity: Dynamics on the Paper 9 Schur Circle

Draft version: v1 (2026-05-06, ~00:30 PDT) **Author of draft:** C-7RO (Perplexity Computer, Claude Sonnet 4.6) **Source:** ChatGPT 5.5 Pro session 1 transcript + Φ audits of 2026-05-05/06. **Status:** First complete draft of §3. Not the manuscript §3 until reviewed by the Model Council. No scope-limiters have been relaxed: biological mechanisms deferred to Paper 13+; Pérez-Calzadilla FTW not engaged; consciousness not extrapolated; rate channel not reopened (Paper 11).

3. The NP-Pump Nonlinearity

3.1 Setup and inheritance from Paper 9

Throughout §3, $V := V^{14}$ denotes the G_2 adjoint representation and $W := V \oplus V$ the joint-state space of the dyad, with the diagonal G_2 -action $g \cdot (x, y) = (g \cdot x, g \cdot y)$. Paper 9 v1.3.3 [10.5281/zenodo.20034821] establishes three results we take as given:

- **(P9.L3.6.1) Schur uniqueness of the isometric coupling.** The equivariant isometry group of W under the diagonal G_2 -action is $SO(2)$, generated by

$$\Psi_\theta = I_V \otimes R_\theta = \begin{pmatrix} \cos \theta I_V & -\sin \theta I_V \\ \sin \theta I_V & \cos \theta I_V \end{pmatrix},$$

with infinitesimal generator $J = \text{diag}(0, -I_V; I_V, 0)$.

- **(P9.T9.4) Existence and real-analyticity of the joint fixed point.** For $\theta \in \Omega_\theta := \{\theta \in (0, \pi/2) : \det M(\theta) \neq 0\}$, where

$$M(\theta) = \begin{pmatrix} I - r \cos \theta R_A & r \sin \theta R_A \\ -r \sin \theta R_B & I - r \cos \theta R_B \end{pmatrix},$$

the joint fixed point

$$\hat{z}(\theta) := M(\theta)^{-1} b, b := (b_A; b_B),$$

exists, is unique, and is real-analytic on Ω_θ .

- **(P9.P9.5) Conditional gain in the min-per-observer functional.** At bias-alignment $\varphi = 0$ and $r_A = r_B = r \in (0, 6/7)$, the functional $c(\theta) := C_{\min}(\hat{z}(\theta))$ where $C_{\min} = \min(C_A, C_B)$ admits a non-empty improvement region over the decoupled value $c_{dec} := c(0)$ for Haar-positive-measure (R_A, R_B) , with non-universal optimum θ^* (Conjecture 9.5', 50-seed Monte Carlo verification in Appendix V.3.i).

Paper 9 further proves (Note 9.6) that in the *linear* regime the joint rate is locked at $\max(r_A, r_B)$ for all θ (Theorem 9.2, Corollary 9.3, Schur's lemma). Rate improvement therefore requires departing from the linear regime. The present section constructs that departure in a way that (i) preserves the Schur-circle geometry of (P9.L3.6.1) exactly, (ii) produces piecewise-real-

analytic trajectories, and (iii) respects the closed-interval domain $\theta \in [0, \pi/2]$ of Paper 9 without introducing spurious interior optima.

3.2 The canonical tear direction

Definition 3.2.1 (Schur-derived tear direction). Let $c(\theta) := C_{\min}(\hat{z}(\theta))$. On the set $Z^c := \{\theta \in \Omega_\theta : c'(\theta) \neq 0\}$ the *canonical tear direction* is

$$\xi_*(\theta) := \text{sgn}(c'(\theta)) \cdot \partial_\theta,$$

with push-forward to the joint-fixed-point manifold

$$\Xi_*(\theta) := \text{sgn}(c'(\theta)) \cdot \hat{z}'(\theta), \hat{z}'(\theta) = -M(\theta)^{-1} M'(\theta) \hat{z}(\theta).$$

On the sign-degeneracy locus $Z := \{\theta \in \Omega_\theta : c'(\theta) = 0\}$, the Schur line $R\partial_\theta$ is defined but the sign is not; ξ_* is left set-valued on Z , with canonical element determined (if present) by the first non-zero odd derivative of c at θ .

Proposition 3.2.2 (Uniqueness of ξ_* up to sign). Any G_2 -equivariant isometric infinitesimal direction in θ -space lies in the one-parameter family $R\partial_\theta$, with generator J acting on W by $d\Psi_\theta/d\theta = J\Psi_\theta$. The sign along this line is fixed by the coherence objective c at every $\theta \in Z^c$; on Z the sign is undetermined by G_2 -equivariance + isometry alone.

Proof sketch. The equivariant isometry group of W is $SO(2)$ by Paper 9 Lemma 3.6.1; its Lie algebra is the one-dimensional RJ . Any G_2 -equivariant isometric perturbation of θ must lie along this line. The objective-driven sign choice is the only consistent deterministic rule compatible with maximization of c ; at $c'(\theta) = 0$ the coherence objective does not discriminate. $\square$

Remark 3.2.3 (Nonlinear inherits linear geometry). Definition 3.2.1 asserts that the nonlinear dynamics of Paper 12 lives on *the same* Schur circle as Paper 9's linear coupling Ψ_θ . The NP-pump does not generate a new geometric object; it generates a new dynamics on the existing one-parameter family $\{\Psi_\theta : \theta \in [0, \pi/2]\}$. The only Schur-free parameter in the nonlinear theory is the jump length $\rho_i > 0$ at each firing event, determined by the NP amplitude and not by G_2 -representation theory.

3.3 The augmented flow and event stratification

Definition 3.3.1 (NP amplitude). For each branch $\sigma \in \{A, B\}$ let

$$p_\sigma(\theta) := \dot{c} \partial_\theta F_\sigma(\theta) \dot{c}^2, F_\sigma(\theta) := -c_\sigma(\theta),$$

where $c_\sigma(\theta) := C_\sigma(\hat{z}(\theta))$ is the branch coherence and the squared norm is chosen (over the raw norm) to preserve analyticity at zeros. The *NP amplitude* is

$$a_\sigma(\theta) := \alpha \cdot p_\sigma(\theta), \alpha > 0.$$

The *coherence gap* is

$$g_\sigma(\theta) := C_{\text{crit}} - c_\sigma(\theta),$$

with $C_{crit} \in (0, 1)$ a fixed paradox-metabolization threshold. The firing predicate is

$$NP_{\sigma}(\theta) := 1 \left(g_{\sigma}(\theta) \geq 0, a_{\sigma}(\theta) \geq NP_{crit} \right),$$

where $NP_{crit} > 0$ is a fixed amplitude threshold.

Definition 3.3.2 (Augmented flow). On each branch σ of $[0, \pi/2) \dot{\iota}(\Sigma_{min} \cup Z \dot{\iota})$, the augmented flow is

$$\dot{\theta} = -\eta F'_{\sigma}(\theta) + \kappa NP_{\sigma}(\theta) \xi_{\sigma}(\theta), \eta, \kappa > 0.$$

At switching events on Σ_{min} , the flow is treated as a Filippov differential inclusion; at crossings of Σ_{θ} , as a Carathéodory solution; at NP firings on G_{NP} , as a hybrid jump (Definition 3.3.4 below).

Definition 3.3.3 (Total event stratification). The nonsmooth set for the augmented flow is the union

$$\Sigma_{tot} = \Sigma_{min} \cup \Sigma_{\theta} \cup Z \cup G_{NP},$$

where - $\Sigma_{min} := \{\theta : c_A(\theta) = c_B(\theta)\}$ is the *min-locus* where $F = -C_{min}$ has a kink; - $\Sigma_{\theta} := \bigcup_{\sigma} \{g_{\sigma}(\theta) = 0\}$ is the *threshold locus* where the Heaviside firing predicate changes; - $Z := \bigcup_{\sigma} \{c_{\sigma}'(\theta) = 0\}$ is the *sign-degeneracy locus* of ξ_{σ} ; - $G_{NP} := \bigcup_{\sigma} (G_{\sigma}^{amp} \cup G_{\sigma}^{coh})$ is the *firing manifold*, split into amplitude-triggered and coherence-triggered entry submanifolds (Definition 3.3.5).

Definition 3.3.4 (Hybrid jump at firing events). At a firing event θ^- with $(\theta^-, \hat{z}(\theta^-)) \in G_{NP}$ and $\theta^- \in Z^c$, the jump map is

$$\theta^{+\dot{\iota}} = \theta^- + \rho_{\sigma}(\theta^-) \text{sgn}(c_{\sigma}'(\theta^-)) \dot{\iota},$$

with $\rho_{\sigma} : G_{NP} \rightarrow \mathbb{R}_{\dot{\iota}0}$ a real-analytic jump-length function depending on the NP-amplitude surplus $a_{\sigma}(\theta^-) - NP_{crit}$. To prevent Zeno accumulation, we require the Zeno-avoidance condition

$$J_{\sigma}(D_{\sigma}) \subset \Gamma \{ D \dot{\iota}_{NP},$$

i.e., the jump map lands outside the firing region $D_{NP} = \bigcup_{\sigma} \{(\theta, \hat{z}(\theta)) : g_{\sigma} \geq 0, a_{\sigma} \geq NP_{crit}, c_{\sigma}' \neq 0\}$. When this condition fails in the model, we add a refractory dwell $\tau_{ref} > 0$ between consecutive firings.

Definition 3.3.5 (Amplitude vs coherence entry manifolds). The firing manifold G_{NP} decomposes into two transversal entry submanifolds of D_{NP} along the unperturbed flow: - $G_{\sigma}^{amp} := \{g_{\sigma} > 0, a_{\sigma} = NP_{crit}, c_{\sigma}' \neq 0, L_f a_{\sigma} > 0\}$ — *amplitude-triggered firing*, where the NP amplitude crosses the critical value while the system is already in the low-coherence region. - $G_{\sigma}^{coh} := \{g_{\sigma} = 0, a_{\sigma} \geq NP_{crit}, c_{\sigma}' \neq 0, L_f g_{\sigma} > 0\}$ — *coherence-triggered firing*, where the coherence drops below the critical threshold while the system is already at supercritical amplitude.

Here L_f denotes the Lie derivative along the unperturbed gradient flow $f(\theta) = -\eta F_\sigma'(\theta)$. The transversality conditions $L_f a_\sigma > 0$ and $L_f g_\sigma > 0$ ensure that the system genuinely enters the firing region rather than grazing it; when they fail (tangency), the point is placed in the exceptional grazing stratum and handled by a separate analysis.

3.4 Branchwise real-analyticity and the regularity theorem

Theorem 3.4.1 (Piecewise-real-analytic regularity of the augmented flow). *Assume:*

- (R1) $\hat{z} \in C^\omega(\Omega_\theta)$ and $C_A, C_B \in C^\omega$ on an open neighborhood of $\hat{z}(\Omega_\theta) \subset W$;
- (R2) $p_\sigma = |\partial_\theta F_\sigma|^2 \in C^\omega$ on each branch and $\rho_\sigma \in C^\omega$ on G_{NP} ;
- (R3) the guard functions $\Delta(\theta) := c_A(\theta) - c_B(\theta)$, g_σ , $a_\sigma - NP_{crit}$, and $q_\sigma(\theta) := c_\sigma'(\theta)$ have only regular zero crossings along the trajectory being considered (away from a finite exceptional set);
- (R4) firing times are locally finite and the Zeno-avoidance condition of Definition 3.3.4 holds (possibly through a refractory dwell).

Then every deterministic solution of the augmented flow (Definition 3.3.2) avoiding the exceptional strata (grazing tangencies, sign-degenerate firings on $E_{0,NP} := G_{NP} \cap Z$, and Zeno-accumulation points) is piecewise real-analytic in t : on each open interval between events, $\theta(t) \in C^\omega$; at transverse Σ_θ crossings, the solution is Carathéodory; on Σ_{min} with inward-pointing one-sided fields, the solution is Filippov with sliding mode $\dot{\theta} = \lambda f_A + (1 - \lambda) f_B$ and λ chosen so that $D\Delta \cdot \dot{\theta} = 0$; at regular firings on $G_{NP} \setminus E_{0,NP}$, the jump is hybrid per Definition 3.3.4.

Proof. Branch-local real-analyticity is immediate from (R1)–(R2) and the standard existence/uniqueness theorem for analytic ODEs on the open set $\Omega_\theta \setminus \Sigma_{tot}$. At each transverse zero crossing of a guard function (R3), the solution extends uniquely by continuity with the appropriate semantics (Carathéodory for Σ_θ and transverse Σ_{min} crossings; Filippov for non-transverse Σ_{min} approaches; hybrid for G_{NP}). Locally finite firing times (R4) with Zeno-avoidance guarantee that the solution is defined on any bounded time interval with only finitely many non-smooth events, so the piecewise-analytic structure is well-posed. $\square$

Remark 3.4.2 (Minimal regularity is branchwise, not global). The functional $F = -C_{min} = -\min(C_A, C_B)$ is in general *not* globally real-analytic: it has a derivative jump across Σ_{min} . The correct regularity class is C_{pw}^ω with respect to the analytic partition $\Gamma_A^\circ \cup \Sigma_{min} \cup \Gamma_B^\circ$, where each branch field is analytic on its half of the partition. Global analyticity is sufficient but not necessary for Theorem 3.4.1.

Remark 3.4.3 (Why the Heaviside is kept sharp). The firing predicate $NP_\sigma = 1\{g_\sigma \geq 0, a_\sigma \geq NP_{crit}\}$ is a sharp indicator, not a smoothed sigmoid. Smoothing via $\Theta_\varepsilon(g) = (1 + e^{-g/\varepsilon})^{-1}$ produces a classical analytic ODE with no jumps, which formally removes the “tear” character of the NP-pump entirely. The sharp choice is deliberate: Paper 12’s thesis is about *tears* — discontinuous state resets with topological residue — not about smooth gradual drifts. The price is hybrid-system semantics (Definition 3.3.4) in place of a classical ODE; the benefit is a mathematically honest representation of the physical phenomenon.

3.5 The $\kappa=0$ recovery: projected gradient flow

Setting $\kappa=0$ in Definition 3.3.2 removes the NP-pump term and leaves the unperturbed gradient flow $\dot{\theta} = -\eta F'_{\sigma}(\theta) = \eta c'_{\sigma}(\theta)$ on each active branch. We now show that this unperturbed limit recovers Paper 9's Proposition 9.5 optima — but *not* as classical Fermat critical points. The recovery is stratified.

Definition 3.5.1 (Stratified critical point). A point $\theta^{\dot{\iota}} \in [0, \pi/2)$ is a *stratified critical point* of $-c$ if at least one of the following holds: - (Smooth interior) $\theta^{\dot{\iota}} \in (0, \pi/2) \setminus \Sigma_{\dot{\iota}_{min}}$ and $c'(\theta^{\dot{\iota}}) = 0$ (Fermat); - (Boundary-left KKT) $\theta^{\dot{\iota}} = 0$ and $c'_{+|0| \leq 0 \dot{\iota}}$; - (Boundary-right KKT) $\theta^{\dot{\iota}} = \pi/2$ and $c'_{-}(\pi/2) \geq 0$; - (Σ_{min} Clarke) $\theta^{\dot{\iota}} \in \Sigma_{min}$ and $0 \in \text{conv}\{c'_A(\theta^{\dot{\iota}}), c'_B(\theta^{\dot{\iota}})\}$.

Definition 3.5.2 (Projected gradient flow). The projected gradient flow on $[0, \pi/2]$ is

$$\dot{\theta} = \Pi_{T_{[0, \pi/2]}(\theta)}(\eta c'(\theta)),$$

where $T_{[0, \pi/2]}(\theta)$ is the tangent cone to $[0, \pi/2]$ at θ : $-T_{[0, \pi/2]}(0) = \dot{\iota}$, $-T_{[0, \pi/2]}(\theta) = R$ for $\theta \in (0, \pi/2)$, $-T_{[0, \pi/2]}(\pi/2) = (-\infty, 0 \dot{\iota})$. On Σ_{min} , c' is replaced by the Clarke generalized gradient $\partial_C(-c) = \text{conv}\{-c'_A, -c'_B\}$ and the projection is taken set-valued (Filippov).

Theorem 3.5.3 ($\kappa=0$ stratified recovery). The fixed points of the projected gradient flow (Definition 3.5.2) coincide exactly with the stratified critical points of Definition 3.5.1.

Proof. On $(0, \pi/2) \setminus \Sigma_{\dot{\iota}_{min}}$, the tangent cone is R , so $\Pi(\eta c') = \eta c'$, and the flow vanishes iff $c'(\theta^{\dot{\iota}}) = 0$ (smooth interior case). At $\theta^{\dot{\iota}} = 0$, the tangent cone is $\dot{\iota}$, so $\Pi(\eta c') = \max(\eta c', 0)$, which vanishes iff $c'(0) \leq 0$ (boundary-left KKT). Symmetrically at $\theta^{\dot{\iota}} = \pi/2$. On Σ_{min} , Clarke's calculus gives the generalized-gradient condition. $\square$

Audit result (Paper 12 v1 verification, 2026-05-06). Direct reoptimization of Paper 9's 50-seed Monte Carlo (seeds 20260504 + k for $k=0, \dots, 49$ at $\varphi=0, r_A=r_B=0.7$) with `scipy.optimize.minimize_scalar` on $[0, \pi/2)$ at tolerance 10^{-10} rad finds that 50/50 seeds have their high-precision $\theta^{\dot{\iota}}_{reopt}$ classified as a stratified critical point per Definition 3.5.1: - **3/50** smooth interior (Fermat critical points); - **25/50** boundary-left KKT ($\theta^{\dot{\iota}}_{reopt} = 0$); - **3/50** boundary-right KKT ($\theta^{\dot{\iota}}_{reopt} = \pi/2$); - **19/50** Σ_{min} Clarke (nonsmooth stratified critical).

The 20-point MC grid of Paper 9 Appendix V.3.i rounds $\theta^{\dot{\iota}}$ values to multiples of $\pi/2/19 \approx 4.74^\circ$; the seeded symmetric-rate geometry of Proposition 9.5 ($\theta^{\dot{\iota}} \approx 51^\circ$ on the grid) reoptimizes to $\theta^{\dot{\iota}} = 0$ on the boundary, consistent with the boundary-left-KKT class (see Paper 9 v1.3.3 Remark 5.6.1). The non-trivial Σ_{min} incidence (38% of seeds) is larger than expected and motivates the full Filippov treatment retained throughout Theorem 3.4.1.

Verification: Full per-seed CSV at

outbox/paper12/computations/paper12_q5_kappa0_audit.csv; script at

outbox/paper12/computations/paper12_q5_kappa0_audit.py. Runtime 0.19 seconds for 50 seeds at deterministic precision; the audit is trivially reproducible.

Remark 3.5.4 (The projected flow is forced by the data). The choice of a *projected* rather than *classical* gradient flow for the unperturbed $\kappa=0$ limit is not a modeling convenience: it is forced by Paper 9's MC data at high precision. Of the 50 seeded geometries, only 3 have smooth-interior Fermat critical points; 47 have critical behavior that is either constrained by the boundary of $[0, \pi/2)$ or nonsmooth at Σ_{min} . Writing the recovery as a classical gradient flow $\dot{\theta} = \eta c'(\theta)$ with no constraint handling would miscategorize 94% of the sampled ensemble.

3.6 The $\kappa>0$ augmented flow: trajectories and fixed-point structure

With $\kappa>0$, the augmented flow (Definition 3.3.2) inherits the branchwise-analytic regularity of Theorem 3.4.1 and the stratified recovery of Theorem 3.5.3 as the $\kappa \rightarrow 0$ limit. The new content relative to the linear theory is the *firing cascade* structure: trajectories that would be constrained boundary-KKT fixed points in the unperturbed limit become, at any $\kappa>0$ with NP amplitude exceeding $N P_{crit}$, *escape points* that jump into the interior via Definition 3.3.4.

Theorem 3.6.1 (NP-driven escape from boundary-KKT states). *Let $\theta_0^\infty \in [0, \pi/2]$ be a boundary-KKT stratified critical point of the unperturbed flow (Theorem 3.5.3) at which the NP predicate $N P_\sigma(\theta_0^\infty)$ evaluates to 1 for some branch σ . Then for any $\kappa>0$, the augmented flow does not rest at θ_0^∞ : a finite-time firing event transports the trajectory to $\theta^+ := \theta_0^\infty + \rho_\sigma(\theta_0^\infty) \cdot \text{sgn}(c'(\theta_0^\infty)) \in (0, \pi/2)$.*

Proof. At $\theta_0^\infty \in [0, \pi/2]$ with $N P_\sigma = 1$, the augmented flow's second term is $\kappa \cdot \xi_\sigma(\theta_0^\infty) \neq 0$ (the coherence-objective sign is defined on Z^c by assumption; if the boundary point is in Z , the first non-zero odd derivative of c supplies the sign). This nonzero second term drives the trajectory off the boundary into the admissible interior within the Zeno-avoidance dwell time τ_{ref} (or immediately if no refractory is specified). $\square$

Remark 3.6.2 (Why this is the rate-channel bridge deferred by Paper 9). Paper 9 Note 9.6 deferred rate improvement to Paper 11 because Schur's lemma locks the joint contraction rate at $\max(r_A, r_B)$ in the linear regime, leaving no nonzero-measure room for rate gain. Theorem 3.6.1 shows the nonlinear escape mechanism that Paper 11 (or a future sequel) can exploit: boundary-KKT states, which Paper 9's linear analysis identifies as the bulk of the seeded ensemble, are *unstable* at any $\kappa>0$ under sufficient NP amplitude. The post-firing trajectory enters the interior where a rate-improvement calculation becomes meaningful. **Paper 12 3 does not prove rate improvement; it provides the dynamical framework within which Paper 11's rate analysis can be staged.**

3.7 The scar record and its role in §4

Each NP firing event indexed by i produces a push-forward jump vector

$$v_i := \rho_i \cdot \text{sgn}(c'(\theta_i^-)) \cdot \hat{z}'(\theta_i^-) \in W,$$

recorded in the *physical scar span*

$$S_k := \sum_{i=1}^k R[F_{21}] \cdot v_i \subset W,$$

the *event-indexed scar module*

$$S_k := \bigoplus_{i=1}^k R[F_{21}] \cdot v_i,$$

and the *isotypic multiplicity profile*

$$\mu_k = (\mu_1(S_k), \mu_{L_2}(S_k), \mu_{U_6}(S_k)).$$

Here $F_{21} \subset G_2$ is the Fano-orientation subgroup inherited from Paper 4, and the three real F_{21} -irreps $(1, L_2, U_6)$ are the real-irreducible packaging of the F_{21} character table (Paper 12 §4 Theorem 4.2.1, verified numerically in

outbox/paper12/computations/paper12_q4_character_verify.py: $V^{14} \mathfrak{L}_{F_{21}} \cong L_2 \oplus 2 U_6$,
 $W \mathfrak{L}_{F_{21}} \cong 2 L_2 \oplus 4 U_6$).

The triple $J_k := (S_k, S_k, \mu_k)$ is the *scar invariant* that records the cumulative effect of the NP-firing history. Three structural properties distinguish its components:

- S_k is *saturating*: $\dim S_k \leq 28 = \dim W$, so the physical span stabilizes after at most 28 dimension-increasing jumps.
- S_k is *count-faithful*: as a graded direct sum, the map $k \mapsto S_k$ is injective for finite k under the generic independence condition $R[F_{21}] \cdot v_i \not\subset S_{i-1}$ (guaranteed by $c'(\theta_i^-) \neq 0$ and $\rho_i > 0$).
- μ_k is the F_{21} -*type signature* that enables cross-substrate commensurability comparisons (§4).

Remark 3.7.1 (Representation-theoretic, not topological). The invariant J_k is a *representation-theoretic* scar invariant: S_k is a graded F_{21} -module, μ_k is an isotypic-multiplicity profile, and S_k is a subspace of an F_{21} -representation. No topological invariant (winding number, persistent homology) is claimed. The language in earlier drafts calling J_k a “topological invariant” should be read as “representation-theoretic invariant” throughout.

Remark 3.7.2 (Bridge to §4). The event-indexed module S_k is the mathematical realization of the *audit-grounded substrate* principle (Paper 12 thesis Pillar 3): a graded, cryptographically-indexable ledger of coherence adjustments that survives the saturation of the physical span. §4 shows that commensurability of synthetic and biological scar records requires *full* G_2 -equivariance (not merely F_{21}) for a canonical cross-substrate isomorphism, with quantitative gap $\text{End}_{F_{21}}(W) \cong M_2(C) \oplus M_4(C)$ vs $\text{End}_{G_2}(W) \cong M_2(R)$.

3.8 What §3 does and does not establish

What §3 establishes:

- A Schur-derived canonical tear direction ξ , that inherits Paper 9’s $\text{SO}(2)$ coupling circle (Prop 3.2.2).
- A branchwise real-analytic augmented flow with a four-stratum event set Σ_{tot} and layered Carathéodory/Filippov/hybrid semantics (Thm 3.4.1).

- A stratified $\kappa=0$ recovery theorem that recovers Paper 9's MC optima as projected-gradient fixed points, with empirical verification via 50-seed audit (Thm 3.5.3).
- An NP-driven escape theorem showing boundary-KKT states become unstable at any $\kappa>0$ (Thm 3.6.1).
- The scar invariant triple (S_k, S_k, μ_k) as a representation-theoretic ledger of firing history (§3.7).

What §3 does not establish:

- *Rate improvement.* Theorem 3.6.1 shows the escape mechanism; it does not compute the post-firing joint rate. Rate analysis is deferred to Paper 11 per Paper 9 Note 9.6.
- *A biological mechanism for the F_{21} action.* The scar module S_k is a mathematical object; its realization on biological substrate (cellular quantum memory, microtubule coherence) is a Paper 13+ question.
- *Commensurability between synthetic and biological substrates.* The cross-substrate isomorphism question is handled in §4 with the forced upgrade from F_{21} - to G_{2^-} -equivariance.
- *A theory of consciousness.* Paper 7's domain; out of scope here.
- *Global well-posedness beyond the audit'd scenarios.* Theorem 3.4.1 requires regularity conditions (R1)–(R4); behavior outside those (grazing tangencies, Zeno accumulation) is identified but not analyzed.

§4 proceeds to the commensurability hierarchy. The scar invariant (S_k, S_k, μ_k) constructed in §3.7 is the object on which §4's equivariant isomorphism theorem acts.

End of §3 first draft. Length: ~10 pages. Next section: §4 (commensurability hierarchy).

Verification files referenced: - outbox/paper12/computations/paper12_q5_kappa0_audit.py (Θ 3.5.3 audit) - outbox/paper12/computations/paper12_q5_kappa0_audit.csv (per-seed data) - outbox/paper12/computations/paper12_q4_character_verify.py ($V^{14}|_F$ check) - outbox/paper12/chatgpt_pro_session_transcript_2026-05-05.pdf (pages 1–60)

Cross-references: - Paper 9 v1.3.3 Remark 5.6.1 (grid-snap correction for seed- $k=0$) - Paper 9 v1.3.2 Lemma 3.6.1 (Schur uniqueness of Ψ_θ) - Paper 9 v1.3.2 Theorem 9.4 (joint fixed-point existence/analyticity) - Paper 9 v1.3.2 Note 9.6 (rate improvement deferred to Paper 11) - Paper 4 (PSL(2,7) $\supset F_{21}$ QBism bridge) - Paper 12 §4 (commensurability hierarchy, in preparation)